



Operating Instructions

Diesel engine

16V 1163 TB73L

20V 1163 TB73L

MS15017/04E

Engine model	kW/cyl.	Application group
16V1163TB73L	325 kW/cyl.	1B; continuous operation, variable, high load
20V1163TB73L	325 kW/cyl.	1B; continuous operation, variable, high load

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

© Copyright Rolls-Royce Solutions

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of Rolls-Royce Solutions. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage and/or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. Rolls-Royce Solutions reserves the right to change, delete or supplement the information provided as and when required.

California Proposition 65

This warning applies only to the State of California, USA, in compliance with Title 27 California Code of Regulations, Article 6, Clear and Reasonable Warnings.



WARNING: Breathing diesel engine exhaust exposes you to chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

- Always start and operate the engine in a well-ventilated area.
- If in an enclosed area, vent the exhaust to the outside.
- Do not modify or tamper with the exhaust system.
- Do not idle the engine except as necessary.

For more information go to www.P65warnings.ca.gov/diesel.

(→ www.P65warnings.ca.gov/diesel)

Table of Contents

1	Safety		5.11	Pump control - Control and display elements	55
1.1	Important provisions for all products	8	5.12	Engine - Prelubrication	56
1.2	Intended use of all products	9	5.13	Plant - Cleaning	57
1.3	Personnel and organizational requirements	10			
1.4	Safety regulations for commissioning and operation	11	6	Maintenance	
1.5	Safety regulations for assembly, maintenance, and repair work	13	6.1	50-hours service	59
1.6	Fire and environmental protection, indirect materials	17	6.2	Maintenance task reference table [QL1]	60
1.7	Standards for warning messages in the text and highlighted information	20	7	Troubleshooting	
2	Transport		7.1	Troubleshooting	62
2.1	Transport	21	7.2	Oil priming pump, oil refill pump - Troubleshooting	71
2.2	Crankshaft transport locking device	22	7.3	Fault indications and fault messages	74
3	General Information		8	Task Description	
3.1	Engine side and cylinder designations	26	8.1	SOLAS	120
3.2	Engine overview	27	8.1.1	SOLAS shields on 16V engines - Installation position	120
3.3	Sensors and actuators	28	8.1.2	Installation locations for SOLAS shielding 20V	163
4	Technical Data		8.1.3	Adhesive tape for SOLAS shielding - Application	206
4.1	Engine data 16V1163TB73L	33	8.1.4	SOLAS shielding as per MTN 5233 - Installation on 16V engines	207
4.2	Engine data 20V1163TB73L	38	8.1.5	SOLAS shielding as per MTN 5233 - Installation on 20V engine	208
4.3	Firing order	43	8.2	Engine	209
4.4	Engine - Main dimensions	44	8.2.1	Engine barring - Manually	209
5	Operation		8.2.2	Cranking engine on starting system	211
5.1	Pump control - Control and display elements	45	8.3	Cylinder Liner	212
5.2	Putting the engine into operation after extended out-of-service periods (>3 months)	46	8.3.1	Combustion chambers - Draining condensate	212
5.3	Preparations for putting into operation after scheduled out-of-service-period	47	8.3.2	Cylinder liner - Endoscopic examination	213
5.4	Regeneration burn-off routine after low-load operation	48	8.3.3	Cylinder liner - Instructions and comments on endoscopic and visual examination	215
5.5	Starting the engine via the automation system	49	8.3.4	Decompression valve - Opening	217
5.6	Operational monitoring	50	8.3.5	Decompression valve - Closing	218
5.7	Engine shutdown via the automation system	51	8.4	Valve Drive	219
5.8	Emergency engine shutdown via automation system	52	8.4.1	Valve clearance - Check and adjustment	219
5.9	After stopping the engine	53	8.4.2	Cylinder head cover - Removal and installation	221
5.10	PPC, LPU - Functional check	54	8.5	Injection Valve / Injector	222
			8.5.1	Injector - Cleaning	222
			8.5.2	Injector - Check	224
			8.5.3	Injector - Disassembly	226
			8.5.4	Injector - Assembly	227
			8.5.5	Injector - Replacement	229
			8.5.6	Injector - Hole-type nozzle replacement	230
			8.5.7	Injector - Removal and installation	231

8.6	Fuel System	235	8.14.6	Oil priming pump - Checking direction of rotation	279
8.6.1	Fuel system - Diagram	235	8.15	Oil Filtration / Cooling	280
8.6.2	Leak fuel tank - Draining	236	8.15.1	Disk-pack filter - Cleaning	280
8.6.3	Fuel system - Venting	237	8.15.2	Engine oil filter - Edge-type filter element cleaning	282
8.7	Fuel Filter	238	8.15.3	Engine oil filter - Replacement	283
8.7.1	Fuel filter - Replacement	238	8.15.4	Piston cooling-oil filter - Replacement	284
8.7.2	Fuel prefilter - Differential pressure indicator check and adjustment	241	8.15.5	Cleaning centrifugal oil filter and replacing filter sleeve	286
8.7.3	Fuel prefilter - Draining	242	8.15.6	Centrifugal oil filter bracket - Removal and installation	289
8.7.4	Fuel prefilter - Flushing	243	8.15.7	Engine oil bypass filter - Replacement	290
8.7.5	Fuel prefilter - Filter element replacement	245	8.16	Coolant Circuit, General, High-Temperature Circuit	292
8.8	Preheater	248	8.16.1	Coolant system 16V/20V - Diagram	292
8.8.1	Preheater	248	8.16.2	Drain and venting points	293
8.8.2	Preheater - Function and leak check	250	8.16.3	Vent line - Obstruction check	297
8.8.3	Preheater - Checking heating element(s) (electrical)	251	8.16.4	Heat exchanger - Venting	298
8.9	Charge-Air Cooling General, Left-Hand Side	252	8.16.5	Engine coolant level - Check	299
8.9.1	Intercooler - Drain line check	252	8.16.6	Engine coolant - Change	300
8.9.2	Intercooler - Checking drain line for coolant discharge and obstruction	253	8.16.7	Draining engine coolant	301
8.10	Air Filter	254	8.16.8	Engine coolant - Filling	302
8.10.1	Air filter element and dust bowl - Cleaning	254	8.16.9	Engine coolant pump - Relief bore check	303
8.10.2	Air filter - Replacement	255	8.16.10	Engine coolant - Sample extraction and analysis	304
8.10.3	Air filter - Removal and installation	256	8.16.11	Coolant system - Strainer removal	305
8.11	Air Intake	257	8.17	Coupling	306
8.11.1	Contamination indicator - Signal ring position check	257	8.17.1	Coupling - Condition check	306
8.11.2	Emergency air shut-off flap - Lubrication	258	8.18	Raw Water Pump with Connections	307
8.11.3	Emergency-air shut-off flap - Check operation with electric actuation	259	8.18.1	Raw water system - Diagram	307
8.11.4	Emergency-air shutoff flap - Manual closing/opening	260	8.18.2	Raw water pump - Relief bore check	308
8.12	Exhaust Flap with Actuator	261	8.18.3	Automatic seawater inductor - Function check	309
8.12.1	Supercharging and flap control - Diagram	261	8.19	Engine Mounting / Support	310
8.12.2	Exhaust turbocharger control linkage - Lubrication	263	8.19.1	Engine mounting - Cleaning	310
8.12.3	Air and exhaust flaps - Functional check	264	8.19.2	Engine mounting - Check and adjustment	311
8.13	Starting Equipment	265	8.19.3	Engine mounting - Checking securing screws for firm seating	313
8.13.1	Pressure reducing station for control air - Putting into operation	265	8.20	Wiring (General) for Engine/Gearbox/Unit	314
8.13.2	Pressure reducing station for control air - Venting and shutdown	267	8.20.1	Engine cabling - Check	314
8.13.3	Pressure reducing station for control air - Filter cleaning	268	8.21	Accessories for (Electronic) Engine Governor / Control System	315
8.13.4	Compressed-air tank - Draining condensate	269	8.21.1	ECS - Connectors and housing cleaning	315
8.13.5	Starter - Condition check	270	8.21.2	Plug connections - Check	316
8.13.6	Start pilot - Filling cold-start fluid	271	8.21.3	ECS - Removal and installation	317
8.14	Lube Oil System, Lube Oil Circuit	272	8.22	Control and Display Elements, General	319
8.14.1	Lube oil system - Diagram	272	8.22.1	Limit switch - Check	319
8.14.2	Engine oil level - Check	273	8.22.2	Indicator lamp - LED replacement	320
8.14.3	Engine oil - Change	275	8.22.3	Fluid level monitor - Check	322
8.14.4	Engine oil - Sample extraction and analysis	277	8.23	Emergency Instrumentation (Local Operating Panel)	323
8.14.5	Oil sludge - Check for metallic residues	278	8.23.1	Engine control system - Operational tests	323

9 Appendix A		9.3 Contact person/Service partner	341
9.1 Tightening specifications for screws, nuts and bolts	325	10 Appendix B	
9.2 Abbreviations	339	10.1 Standard Tools and Special Tools	342
		10.2 Index	349

1 Safety

1.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at:<http://www.mtu-solutions.com>

1.2 Intended use of all products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

The product must not be operated in explosive atmospheres unless the engine fulfills the conditions for such use and approval has been granted.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

1.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.